Symbolic Solver Execution and Output Examples

This section provides examples of possible outputs generated by each solver.

For the **Clingo** solver, the possible outcomes are:

- *satisfiable*: A model satisfying the rules was found.
- *optimum found*: The best model based on optimization criteria was found.
- *unsatisfiable*: No model satisfies the rules.
- *unknown*: The solver cannot determine the result conclusively.
- *timeout*: The solver exceeded the time limit.
- *grounding timeout*: The ASP code grounding process exceeded the time limit.

```
"status":"satisfiable",
"command":"clingo dreadsbury.lp 10 --outf=2 --time-limit
    =60",
"models":[
  ["killer_name(\"Agatha\")","killer(0)"]
]
```

Listing 1: Clingo output: satisfiable

```
"status":"optimum_found",
"command":"clingo graph_coloring.lp 10 --outf=2 --time-
    limit=60",
"models":[
  ["assigned(1,4)","assigned(2,5)","num_colors(3)"]
]
```

Listing 2: Clingo output: optimum found

```
"status":"unsatisfiable", "command":"clingo reading_club.
    lp...", "models":[]
```

Listing 3: Clingo output: unsatisfiable

```
"status":"unknown", "command":"clingo scheduling.lp...",
    "models":[]
```

Listing 4: Clingo output: unknown

```
"status":"timeout", "error": "Clingo exceeded 60s"
```

Listing 5: Clingo output: timeout

```
"status":"grounding_timeout",
"error":"CRITICAL ERROR: Grounding timed out (>15s). Your
    ASP program is too large..."
```

Listing 6: Clingo output: grounding timeout

For the **Z3** solver, the possible outcomes are:

- *sat*: A satisfying model was found.
- *unsat*: No satisfying model exists.
- *proved*: A definitive answer was reached via theorem proving.
- *unknown*: The solver cannot determine the result.

```
"stdout": "STATUS: sat\\r\\nanswer:C\\r\\n"
```

Listing 7: Z3 output: sat

```
"stdout": "STATUS: unsat\\r\\n Refine: No Options found"
```

Listing 8: Z3 output: unsat

```
"stdout": "STATUS: proved\\r\\ Conclusion : \"TRUE\" \\r
   \\n"
```

Listing 9: Z3 output: proved

```
"stdout": "STATUS: unknown\\r\\n"
```

Listing 10: Z3 output: unknown

For the **Vampire** solver, the possible outcomes are:

- *Theorem*: A refutation was found (conjecture is true).
- *Countersatisfiable*: No refutation was found (conjecture is false).
- *Contradictory Axiom*: There is a contradiction in the given axioms.
- *Syntax Error*: There is a syntax error in the TPTP code.
- *Timeout*: The solver exceeded the time limit.

```
"positive":{"status":"success","szs_status":"Theorem","
   stdout":"% Refutation found..."}
```

Listing 11: Vampire output: Theorem

```
"negative":{"status":"success","szs_status":"
   CounterSatisfiable","stdout":"% SZS status
   CounterSatisfiable..."}
```

Listing 12: Vampire output: Countersatisfiable

```
{"status":"error","szs_status":"Unknown","stdout":"% Time
    limit reached!..."}
```

Listing 13: Vampire output: Timeout

```
"status":"error","szs_status":"Unknown","stdout":"%
   Failed with\\n% User error: Failed to create
   predicate..."}
```

Listing 14: Vampire output: Syntax Error

```
"positive":{"status":"success","szs_status":"
   ContradictoryAxioms","stdout":"% Refutation found...
   ContradictoryAxioms..."}
```

Listing 15: Vampire output: Contradictory Axiom

If the **Language Model** makes an error during an MCP Tool call (e.g., missing arguments), it generates an error message like this:

```
Error executing tool write_and_run_z3: 1 validation error
   ... Missing required argument...
```

Listing 16: Language model MCP Tool call error

This error message is then used for iterative refinement to correct the code and re-execute the tool properly.

Lastly, the language model can experience **overthinking**—producing excessively long internal reasoning that exceeds token limits, leading to an error:

```
[OVERTHINKING] Output tokens (32,768) exceeded threshold
   (32,000). LLM failed Translation...
```

Listing 17: Language model overthinking error

Further analysis of overthinking and the initial evaluation of translation and solver execution will be discussed in the next sections.